Netflix Data Analysis Using Python

Internship Project Report

Project Title Netflix Data Analysis Using Python

Tasks Completed

- Task 1 – Data Cleaning & Preparation
- Task 2 – Content Type Analysis Dashboard
- Task 3 – Country-wise Content Analysis
- Task 4 – Release Year Trend Analysis
- Task 5 – Rating & Genre Analysis
- Task 6 – Business Insights Report

Tools & Technologies

- Python
- Pandas
- Matplotlib
- Jupyter Notebook
- Visual Studio Code

Internship Duration

July 2026 – August 2026

Submitted To	**Submitted by**
Auspify Technologies	**Deepa . K**

Acknowledgement

I would like to express my sincere gratitude to **Auspify Technologies** for providing me with the opportunity to complete this Data Analysis internship.

During this internship, I gained practical experience in data cleaning, exploratory data analysis (EDA), data visualization, and business insight generation using Python. Working on the Netflix dataset helped me understand how real-world datasets can be analyzed to support business decision-making.

I would also like to thank my mentors and everyone who supported me throughout this internship. The guidance and learning experience have significantly improved my technical and analytical skills.

Finally, I am grateful for the opportunity to apply theoretical knowledge to practical projects, which has strengthened my confidence in working with data analysis tools and techniques.

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Chapter 5: Introduction

Introduction

Data has become one of the most valuable assets for organizations across industries. Data analysis enables businesses to discover patterns, identify trends, and make informed decisions based on evidence rather than assumptions.

Netflix is one of the world's leading streaming platforms, offering thousands of movies and television shows across different countries, genres, and audience categories. Analyzing Netflix's content library provides valuable insights into its content strategy, audience preferences, and global expansion.

This internship project focuses on analyzing the Netflix dataset using Python. The project includes data cleaning, exploratory data analysis (EDA), visualization, and business insight generation. Through this analysis, we identify trends in content distribution, release years, ratings, genres, and country contributions, demonstrating how data can support strategic business decisions.

Chapter 6: Project Objectives

Project Objectives

The primary objective of this project is to perform Exploratory Data Analysis (EDA) on the Netflix dataset using Python. The project aims to clean the dataset, analyze trends, create visualizations, and generate business insights that can support data-driven decision-making.

Specific Objectives

- Analyze the distribution of Movies and TV Shows.
- Study Netflix content release trends over the years.
- Identify the top countries contributing to Netflix content.
- Analyze content ratings and target audience categories.
- Discover the most popular Netflix genres.
- Create professional dashboards using Matplotlib.
- Generate meaningful business insights from the analysis.

Chapter 7: Dataset Description

Dataset Description

Dataset Name : Netflix Titles Dataset

Source : Public Netflix Dataset (CSV Format)

Dataset Size

Attribute	Value
Total Records	8,790
Total Columns	10

Dataset Columns

Column Name	Description
show_id	Unique ID for each Netflix title
type	Movie or TV Show
title	Name of the content
director	Director of the title
country	Country of origin
date_added	Date added to Netflix
release_year	Original release year
rating	Audience rating (TV-MA, PG, etc.)
duration	Movie duration or TV seasons
listed_in	Genre/category

Dataset Characteristics

- Structured CSV dataset
- Mixed categorical and numerical data
- Suitable for Exploratory Data Analysis
- Contains Movies and TV Shows from multiple countries

Chapter 8: Tools & Technologies

Tools & Technologies Used

Tool	Purpose
Python	Programming Language
Pandas	Data Cleaning and Analysis
Matplotlib	Data Visualization
Jupyter Notebook	Code Development
Visual Studio Code	IDE
CSV Dataset	Data Source

Python Libraries Used

```
import pandas as pd
import matplotlib.pyplot as plt
```

Chapter 9: Methodology

Project Methodology

The project followed a structured data analysis workflow.

Step 1: Data Collection

The Netflix dataset was loaded from a CSV file into a Pandas DataFrame.

Step 2: Data Cleaning

The dataset was inspected for missing values, duplicates, and incorrect data types before analysis.

Step 3: Exploratory Data Analysis (EDA)

Several analyses were performed, including:

- Content distribution
- Release year trends
- Country analysis
- Rating analysis
- Genre analysis

Step 4: Data Visualization

Professional charts and dashboards were created using Matplotlib.

Step 5: Business Insights

Insights were generated from the visualizations to explain Netflix's content strategy and audience preferences.

Chapter 10:

Task 1: Data Cleaning & Preparation

Objective

The objective of Task 1 was to prepare the Netflix dataset for analysis by loading the data, examining its structure, and ensuring it was suitable for further exploration.

Activities Performed

- Imported Pandas and NumPy libraries.
- Loaded the CSV dataset into a DataFrame.
- Displayed the first five records using `head()`.
- Checked dataset dimensions using `shape`.
- Examined column names using `columns`.
- Reviewed data types using `info()`.
- Generated descriptive statistics using `describe()`.
- Counted Movies and TV Shows using `value_counts()`.

Results

- Successfully loaded **8,790** Netflix records.
- Verified **10 columns** in the dataset.
- Confirmed the dataset was ready for analysis.

Business Insight

The dataset contains a balanced mix of descriptive information about Netflix titles, providing a strong foundation for exploratory data analysis and visualization.

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Task 2 – Content Type Analysis Dashboard

Objective

The objective of this task was to analyze the distribution of Movies and TV Shows on Netflix and visualize the proportion of each content type using charts and dashboards.

Activities Performed

- Loaded the cleaned Netflix dataset.
- Counted Movies and TV Shows using `value_counts()`.
- Created a bar chart to compare the number of Movies and TV Shows.
- Created a pie chart to show the percentage distribution.
- Combined both charts into a dashboard.

Results

- Movies: **6,126 (69.7%)**
- TV Shows: **2,664 (30.3%)**

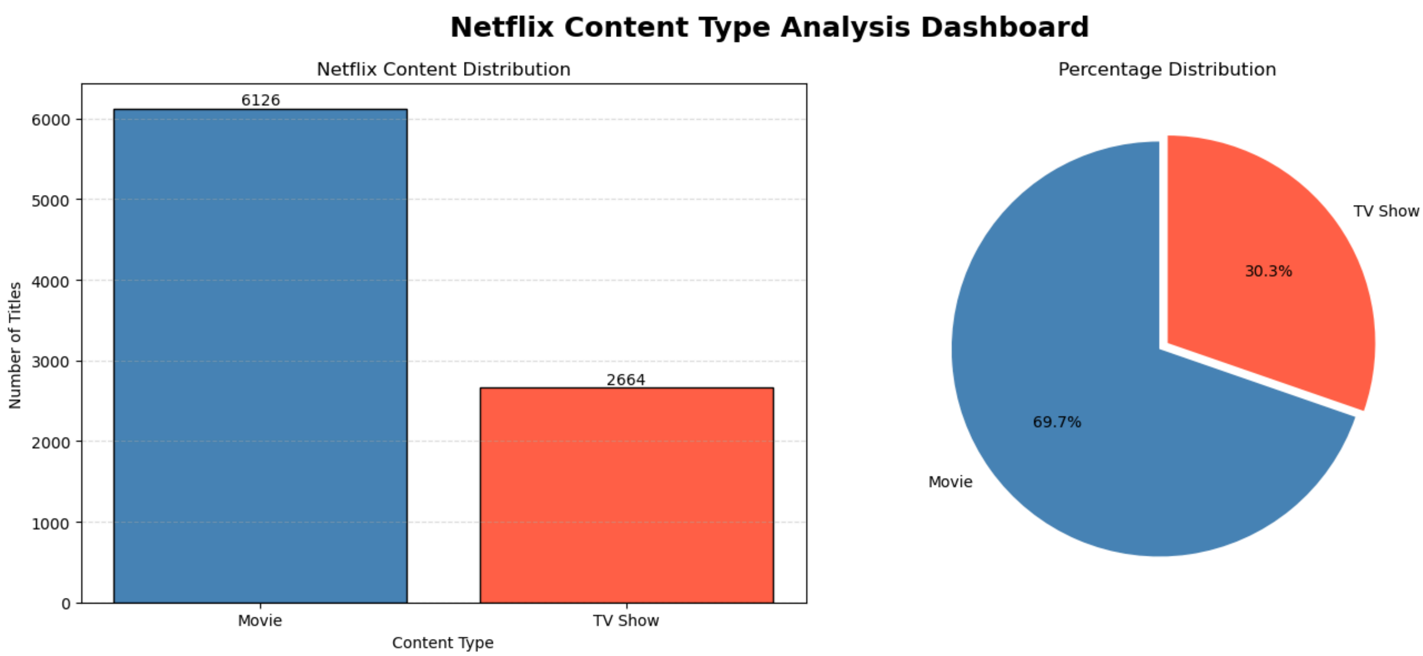
Visualizations

- Bar Chart – Content Distribution
- Pie Chart – Percentage Distribution
- Dashboard – Combined Visualization

Business Insight

Netflix's catalog is dominated by Movies, which account for nearly 70% of all available titles. TV Shows make up approximately 30%, indicating that Netflix places greater emphasis on movies while maintaining a significant television catalog.

Figure 2: Netflix Content Type Dashboard



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Task 3 – Country-wise Content Analysis

Objective

The objective of this task was to identify the countries contributing the most Netflix content and understand the global distribution of titles.

Activities Performed

- Counted the number of titles produced by each country.
- Selected the Top 10 countries.
- Created a horizontal bar chart.
- Calculated the contribution of the Top 10 countries.

Results

Rank	Country	Titles
1	United States	3240
2	India	1057
3	United Kingdom	638
4	Pakistan	421
5	Not Given	287
6	Canada	271
7	Japan	259
8	South Korea	214
9	France	213
10	Spain	182

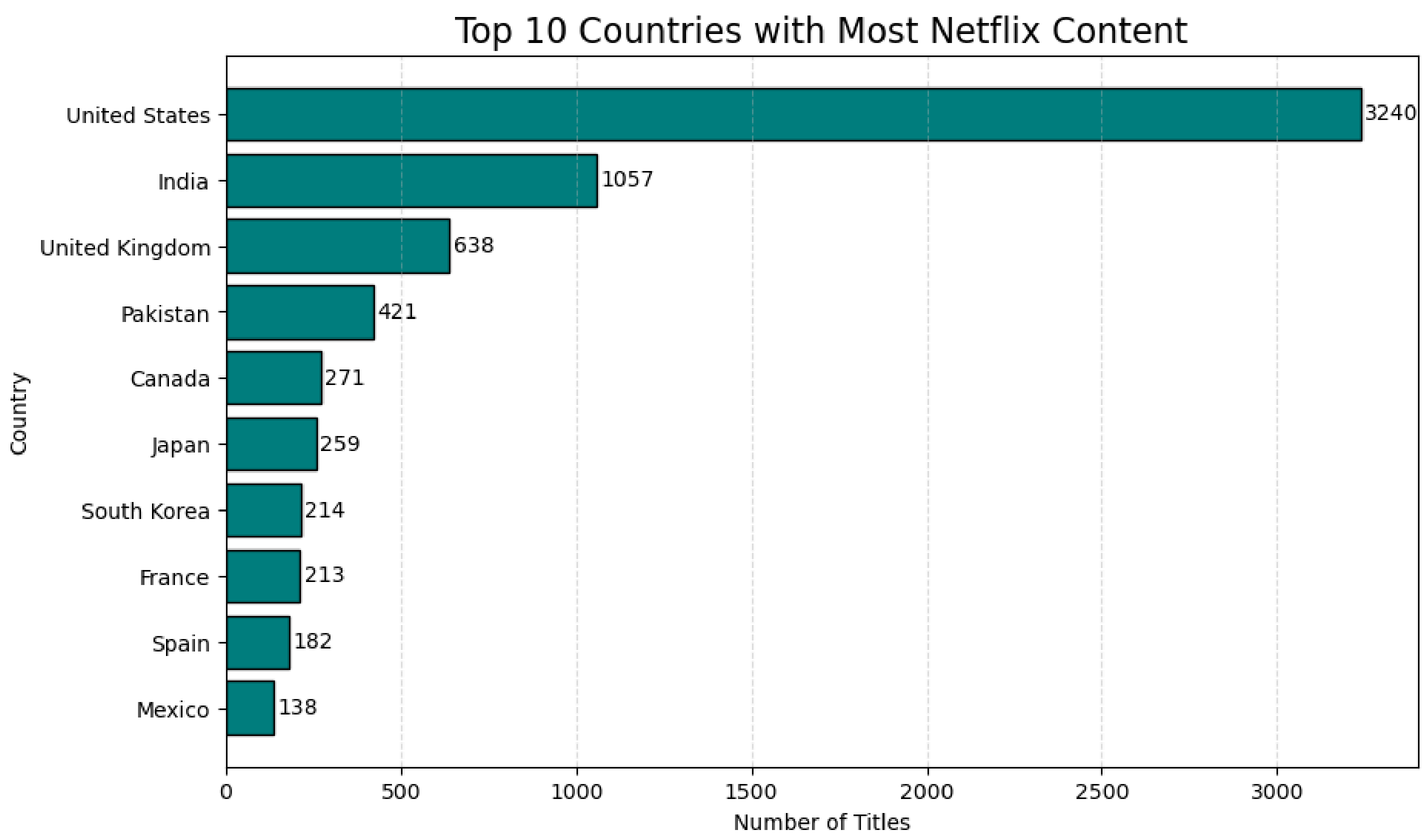
Key Finding

The **Top 10 countries contribute approximately 78.01%** of all Netflix content.

Business Insight

The United States remains Netflix's largest content contributor, while India and the United Kingdom also play significant roles. This reflects Netflix's strong presence in both North America and international markets.

Figure 3: Country-wise Content Analysis



Chapter 13:

Task 4 – Release Year Trend Analysis

Objective

The objective of this task was to analyze how Netflix content production has changed over time.

Activities Performed

- Counted titles released each year.
- Created a line chart to show release trends.
- Identified the Top 10 release years.
- Built a dashboard combining both visualizations.

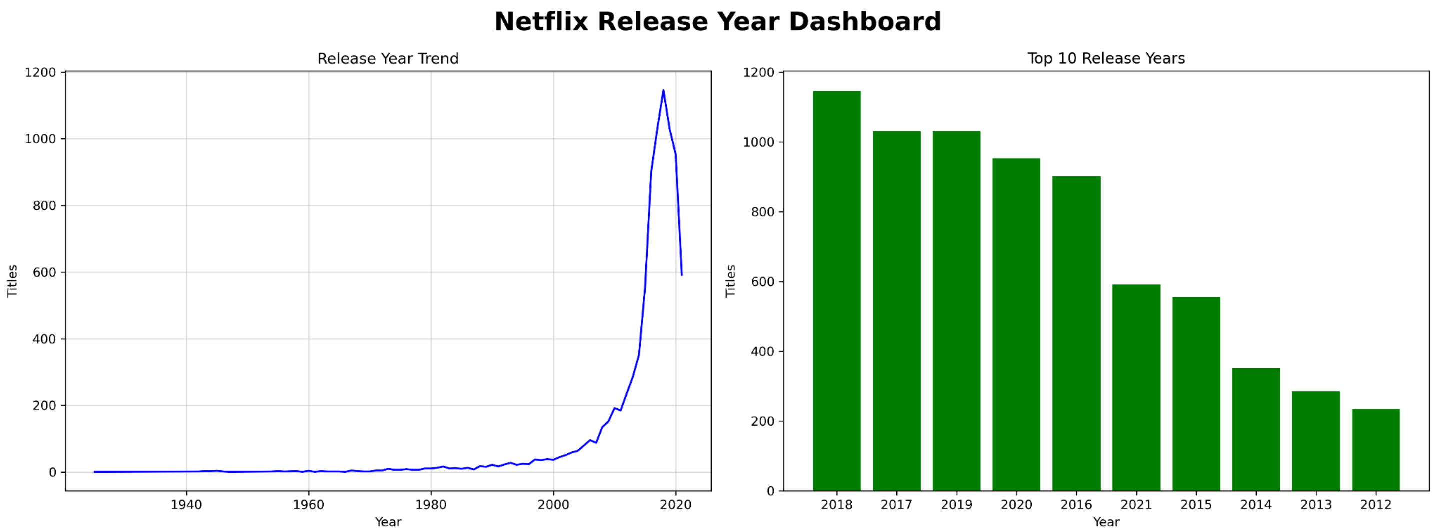
Key Findings

- Content production increased steadily after 2010.
- Rapid growth occurred after 2015.
- Peak production was observed around **2018–2019**.
- A decline is visible after 2020.

Business Insight

Netflix significantly expanded its content production during the late 2010s, reflecting increased investment in original programming and global expansion.

Figure 4: Release Year Dashboard



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Task 5 – Rating & Genre Analysis

Objective

The objective of this task was to analyze Netflix content ratings and genres to understand better audience targeting and content diversity.

Activities Performed

- Counted titles by rating.
- Created a rating distribution chart.
- Analyzed genres using `explode()`.
- Created a Top 10 Genres chart.
- Compared ratings across Movies and TV Shows using `pd.crosstab()`.
- Built a combined dashboard.

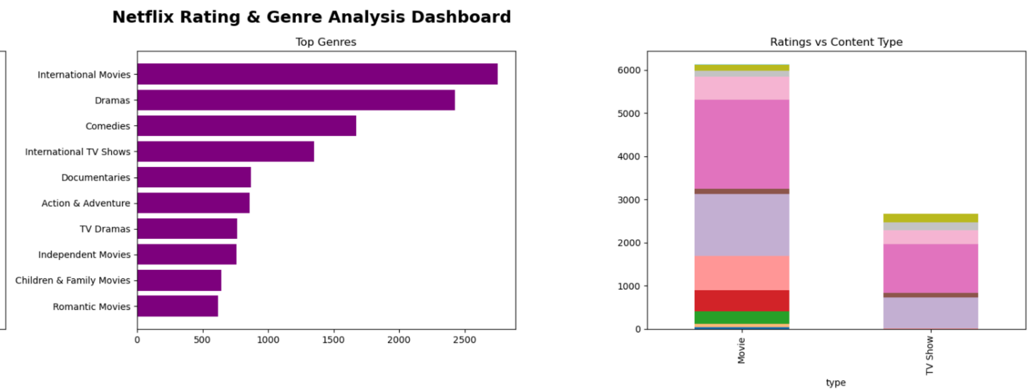
Key Findings

- **TV-MA** is the most common rating.
- **TV-14** is the second most common rating.
- **International movies are** the most popular genre.
- **Dramas** and **Comedies** are also highly represented.

Business Insight

Netflix focuses primarily on mature audiences while maintaining a diverse range of genres to appeal to viewers across different countries and preferences.

Figure 5: Rating & Genre Dashboard



Chapter 15: Task 6 – Business Insights Report

Objective

The objective of this task was to summarize the analysis performed in Tasks 1–5 and present business-oriented insights and recommendations.

Executive Summary

The Netflix dataset was analyzed to understand content distribution, release trends, country contributions, ratings, and genres. The analysis demonstrates how data can be used to support strategic business decisions.

Major Business Insights

- Movies dominate Netflix's content library.
- The United States contributes the highest number of titles.
- The Top 10 countries contribute approximately **78.01%** of all content.
- Netflix experienced significant content growth after 2015.
- Mature audience ratings (TV-MA and TV-14) dominate the platform.
- International Movies are the most common genre.

Recommendations

- Continue investing in international content.
- Maintain a balance between Movies and TV Shows.
- Increase family-friendly programming to broaden audience reach.
- Expand content production in emerging markets.
- Monitor release trends to optimize content planning

Conclusion

The project demonstrates the practical application of Python, Pandas, and Matplotlib for exploratory data analysis and business intelligence. The findings provide meaningful insights into Netflix's content strategy and global distribution.

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Overall Business Insights

Overall Business Insights

The Netflix dataset analysis revealed several important trends about the platform's content strategy, global expansion, and audience preferences.

Content Distribution

Movies make up approximately **69.7%** of Netflix's catalog, while TV Shows account for **30.3%**. This indicates that Netflix has historically focused more on movies while maintaining a strong television content library.

Global Content Distribution

The United States is the largest contributor of Netflix content, followed by India and the United Kingdom. The Top 10 countries account for approximately **78.01%** of all titles, underscoring the importance of a relatively small group of content-producing countries.

Content Growth

Netflix experienced rapid content growth after **2015**, with peak production occurring around **2018–2019**. This period reflects significant investment in original programming and international expansion.

Audience Targeting

The dominance of **TV-MA** and **TV-14** ratings indicates that Netflix primarily targets mature and teenage audiences while still offering family-friendly content.

Genre Trends

International Movies, Dramas, and Comedies are the most common genres. This demonstrates Netflix's strategy of serving a global audience with diverse entertainment options.

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Challenges Faced

Challenges Faced During the Project

During the completion of this project, several practical challenges were encountered and successfully resolved.

Challenge 1: Python Environment Setup

Initially, configuring Python, Jupyter Notebook, and Visual Studio Code required troubleshooting before the development environment was ready.

Challenge 2: Data Cleaning

The dataset contained missing values and duplicate records, which had to be identified and handled before analysis.

Challenge 3: Multi-Genre Analysis

Some Netflix titles belonged to multiple genres. The `explode()` function was used to separate these values for accurate analysis.

Challenge 4: Dashboard Design

Creating clear and readable dashboards required selecting appropriate chart types, colors, labels, and layouts.

Challenge 5: Business Interpretation

Converting charts into meaningful business insights required understanding the data from both a technical and business perspective.

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Learning Outcomes

This internship significantly improved my technical and analytical skills.

Technical Skills

- Python programming
- Data manipulation using Pandas
- Data visualization using Matplotlib
- Exploratory Data Analysis (EDA)
- Dashboard creation
- Business insight generation

Analytical Skills

- Data cleaning
- Trend analysis
- Categorical data analysis
- Comparative analysis
- Data interpretation
- Decision support through visualization

Professional Skills

- Report writing
- Problem-solving
- Logical thinking
- Data-driven decision making
- Documentation
- Presentation preparation

Overall Learning

This project enhanced my confidence in handling real-world datasets and demonstrated how Python can be used to solve business problems through effective data analysis.

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Conclusion

This internship project successfully demonstrated the use of Python for real-world data analysis using the Netflix dataset. The project covered the complete data analysis workflow, including data cleaning, exploratory data analysis (EDA), data visualization, and business insight generation.

Using Pandas, the dataset was cleaned and analyzed to identify patterns in content distribution, release trends, country contributions, audience ratings, and genres. Matplotlib was used to create professional charts and dashboards that made the analysis easier to interpret and communicate.

The analysis revealed that Movies dominate Netflix's content library, the United States is the largest content contributor, and International Movies represent the most common genre. The study also showed that Netflix primarily targets mature audiences through TV-MA and TV-14 rated content while maintaining a diverse catalog for viewers worldwide.

Overall, this internship enhanced my technical skills in Python programming, data analysis, and visualization. It also strengthened my ability to interpret data and convert analytical findings into meaningful business insights. The knowledge and practical experience gained during this project have increased my confidence in working on real-world data analytics projects and prepared me for future opportunities in the field of Data Analytics.

Chapter 20:

References

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